Siddharth Prabhu

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Research Interests

Computer vision and video understanding; real-time and low-latency AI systems; efficient inference and edge deployment of deep networks; multi-object tracking; multimodal document understanding; retrieval-augmented and agentic LLM systems.

My interest sits at the boundary between method and deployment: how detection and tracking models behave under real operating constraints — constrained compute, concurrent streams, long-running unattended processes — and how they can be made measurably faster and more reliable without giving up accuracy.

Education

Siddaganga Institute of Technology 2020 – 2024
B.E. in Electronics and Communication Engineering Tumakuru, Karnataka, India

- Final-year project: vision-based automated detection of motorcycle traffic violations; peer-reviewed and published at IEEE ESCI 2024 (see *Publications*).

Publications

[C1] S. Prabhu, *[co-authors — insert exact author list and order from IEEE Xplore]*, "Image Processing based Traffic Surveillance," *2024 International Conference on Emerging Smart Computing and Informatics (ESCI)*, IEEE, 2024. IEEE Xplore: 10561319

Research Experience

Undergraduate Research Project (Final-Year Thesis) 2023 – 2024
Vision-Based Detection of Motorcycle Traffic Violations Siddaganga Inst. of Technology

- **Problem.** Manual enforcement of two-wheeler traffic law is limited in coverage and inconsistently applied; the project asked whether commodity surveillance footage alone suffices to detect violations and attribute them to a specific vehicle.
- **Dataset.** Co-collected and manually annotated an original dataset of 5,000+ traffic surveillance frames using Roboflow, labelling helmet use, rider count (triple riding) and mobile-phone use during riding.
- **Method.** Trained a custom convolutional network for multi-violation classification, coupled to a downstream licence-plate localisation and character extraction stage so that a detected violation resolves to an identifiable vehicle.
- **Outcome.** Pipeline evaluated on held-out annotated frames; accepted and published in the IEEE ESCI 2024 proceedings, indexed in IEEE Xplore.

Professional Experience

nhance.ai Sept. 2024 – Present
Data Science & AI Engineer Bengaluru, India

- **Edge video analytics.** Built a multi-stream RTSP ingestion and analytics pipeline using YOLOv11 detection with BoT-SORT multi-object tracking. Reduced per-frame inference latency by roughly 40% on CPU-class edge hardware through OpenVINO graph optimisation and model conversion, profiling latency before and after each change.
- **Retrieval-augmented assistant.** Deployed a production RAG system over a large internal corpus of technical documents using Milvus hybrid (dense + sparse) retrieval, treating 95th-percentile query latency rather than mean latency as the governing service metric.
- **Multimodal document intelligence.** Designed workflows using Gemini for summarisation, question answering and structured field extraction across native PDFs and scanned images.
- **Systems work.** Architected an end-to-end MERN configurator automating generation of SOW, BOQ, BRD and proposal documents, replacing a manual multi-hour drafting process. Trained and supervised two interns in edge video analytics and RAG.

M.Tech Solutions Aug. – Sept. 2024
Information Security Intern Bengaluru, India

- Automated threat analysis over security datasets in Python, replacing manual inspection steps in the assessment workflow; implemented Role-Based Access Control (RBAC) to harden access-control infrastructure.

Selected Technical Projects

Real-Time Full-Duplex Voice Agent repository
Python, asyncio, Deepgram STT/TTS, Groq Llama 3.1

- Investigated how to minimise *perceived* end-to-end latency in a cascaded speech → LLM → speech pipeline, where the dominant cost is serialisation between stages rather than single-model inference time.
- Implemented a concurrent triple-loop architecture (listen / think / speak) coordinated by lock-free thread-safe queues, streaming the LLM token stream directly into speech synthesis rather than waiting for full generation.
- Designed an adaptive speech-synthesis flush policy (punctuation-boundary and character-count triggers) trading time-to-first-audio against prosodic continuity, and an RMS noise gate with minimum-utterance filtering to suppress false barge-in — an explicit precision/recall trade-off in voice-activity detection.

Self-Healing Edge Video-Analytics Pipeline repository
YOLOv11, BoT-SORT, OpenCV, Python

- Built a people-counting system as distributed worker processes under a central supervisor that detects crashed or frozen workers and restarts them, targeting unattended long-duration operation — a failure mode rarely addressed in research prototypes.
- Derived directional entry/exit counts from tracker trajectories using line-crossing gate logic over persistent object identities.
- Exposed a live JSON metrics API and dashboard reporting CPU, GPU and memory telemetry alongside throughput, enabling continuous measurement of the operating point; configured via YAML with environment-variable secret handling.

Multimodal Document Question-Answering Service repository
FastAPI, Milvus, Tesseract, Gemini, Docker

- Built a document QA service whose ingestion stage adapts to input modality: native text extraction for digital PDFs, row-wise parsing for spreadsheets, and OCR for image-only documents.
- Exposed `/upload` and `/query` REST endpoints; containerised the full stack with Docker for reproducible deployment.

Technical Skills

Deep learning & vision: PyTorch, TensorFlow, YOLOv8/v11, BoT-SORT, OpenCV, CNNs, RNN/LSTM

Efficient inference & deployment: OpenVINO graph optimisation and model conversion, Docker, multi-process pipelines, latency profiling

LLM & retrieval systems: RAG, hybrid dense/sparse retrieval, Milvus, FAISS, LangChain, agentic patterns (ReAct, multi-agent), Gemini, Llama 3

Languages: Python, JavaScript, SQL, Bash

Systems & backend: FastAPI, asyncio, WebSockets, REST APIs, MongoDB, Node.js, React, microservices; Git, CI/CD, Linux (Fedora / Ubuntu), Roboflow, n8n

Additional Information

- **Technical writing.** Author at algorithm-alchemist.medium.com on retrieval-augmented generation, vector databases and MLOps.
- **Leadership.** Sergeant, National Cadet Corps (NCC) — team coordination and discipline-focused cadet programmes.